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A new proof of the Lagrange multiplier rule

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ABSTRACT

We present an elementary self-contained proof for the Lagrange multiplier rule. It does not refer to any preliminary material and it is only based on the observation that a certain limit is positive. At the end of this note, the power of the Lagrange multiplier rule is analyzed.

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1. Introduction to the Lagrange multiplier rule

The Lagrange multiplier rule gives a method to solve optimization problems with equality constraints. Its value was immediately appreciated when it was discovered. When Euler received in 1755 a letter from the nineteen year old Lagrange communicating new optimization methods (see [16]), he wrote in his answer: "you have brought the theory of optimization to its highest perfection" (see [6]). He withheld publication of his own work on optimization to give the young man the opportunity to work out and publish his method, and receive the honor as the sole discoverer.

Originally, Lagrange had infinite dimensional problems – so-called problems of the calculus of variations – in mind, for applications to classical mechanics, but later he applied his methods also to finite dimensional problems and in 1797 he formulated the multiplier rule in words: "if a function of several variables should be a *maximum* or *minimum* and there are between these variables one or several equations, then it will suffice to add to the proposed function the functions that should be zero, each multiplied by an undetermined quantity, and then to look for the *maximum* or the *minimum* as if the variables were independent; the equations that one will find, combined with the given equations, will serve to determine all the unknowns" (see [10]).

The words of Euler turned out to be prophetic: all methods that have been discovered till the present day for solving optimization problems analytically can be viewed as realizations of Lagrange's sentence above. In particular this holds, in retrospect, for Pontryagin's celebrated maximum principle for optimal control problems (see [17]).

Lagrange never gave a proof of the rule; he restricted himself to applying it with success to many interesting problems. However, others found artificial examples of problems where the rule failed. This puzzling phenomenon was only explained one hundred years after the discovery by Lagrange of the multiplier rule. Then the implicit function theorem was established and this allowed to prove the Lagrange multiplier rule. Here is a precise formulation, where a multiplier is assigned to the objective function as well.

Theorem 1 (*The Lagrange Multiplier Rule*). *Let U be an open set in $\mathbb{R}^n$, the space of n-dimensional column vectors, and f_i, $0 \le i \le m$ be continuously differentiable functions on U. Let $\widehat{x}$ be a point of local minimum of $f_0(x)$ subject to the constraints $f_i(x) = 0$, $1 \le i \le m$. Then the $m+1$ derivatives $f_i'(\widehat{x})$, $0 \le i \le m$, the n-dimensional row vectors of the partial derivatives of the f_i, $0 \le i \le m$ at $\widehat{x}$, are linearly dependent. That is, for suitable numbers (Lagrange multipliers) λ_i, $0 \le i \le m$, not all equal to zero, the Lagrange vector equation holds,*

$$\lambda_0 f_0'(\widehat{x}) + \lambda_1 f_1'(\widehat{x}) + \cdots + \lambda_m f_m'(\widehat{x}) = 0.$$

Remark 1. Here are some remarks about the 'bad case' $\lambda_0 = 0$. Clearly, the case $\lambda_0 = 0$ corresponds to the situation when the derivatives of the constraints $f_i'(\widehat{x})$, $i = 1, \ldots, m$ are linearly dependent. Otherwise, λ_0 is nonzero, hence one can divide by it and get $\lambda_0 = 1$. Thus, *assuming the linear independence of the m*

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vectors $f_i'(\widehat{x})$, $i = 1, \ldots, m$ *we may put* $\lambda_0 = 1$. Thus the conclusion of the theorem – the $m+1$ vectors $f_i'(\widehat{x})$, $0 \le i \le m$ are linearly dependent – is equivalent to $f_0'(\widehat{x}) + \sum_{i=1}^m \lambda_i f_i(\widehat{x}) = 0$ for suitable numbers λ_i, $1 \le i \le m$ under the assumption that the m vectors $f_i'(\widehat{x})$, $1 \le i \le m$ are linearly independent. In applications of the multiplier rule to concrete problems of interest, one can usually show that the case $\lambda_0 = 0$ leads to a contradiction; then one puts $\lambda_0 = 1$. Moreover, in the special case that all constraints are linear, that is, $f_i(x) = a_i x + b_i$, $1 \le i \le m$, one can always choose $\lambda_0 = 1$, even without the linear independence assumption. Indeed, selecting a maximal independent system among the vectors $a_i = f_i'(\widehat{x})$, $i = 1, \ldots, m$ (we call it a basis system), and removing the others we arrive at the same optimization problem. Since the constraints are now independent it follows that one can choose $\lambda_0 = 1$. Now we return to the original problem and put all other multipliers (associated to the non-basis constraints) equal to zero. For some 'artificial' problems, one cannot put $\lambda_0 = 1$; these are counterexamples to the rule as formulated by Lagrange. In the literature, various constraint qualifications (Guignard [8], Abadie [1], Mangasarian–Fromovitz [13]) have been introduced for local solutions $\widehat{x}$—often for the more general necessary conditions for problems where inequality constraints are allowed (the Fritz-John conditions; these generalize the Lagrange vector equation). These constraint qualifications are conditions on $\widehat{x}$ that imply that the Fritz-John conditions hold with $\lambda_0 = 1$.

The Lagrange multiplier rule is one of the most widely known mathematical methods. It would be of interest to all users to be able to read a proof of it, and in particular to understand why there are counterexamples to Lagrange's original formulation and how this formulation can be corrected.

Unfortunately, all existing proofs of the multiplier rule in the literature require effort. Either these proofs refer to substantial basic results or they contain technical arguments. Such substantial basic results are the implicit function theorem [11,12], the inverse function theorem (of which the multiplier rule is an immediate consequence [2]), the tangent space theorem [4,7], the Brouwer fixed-point theorem [9] or the Ekeland variational principle [5]. Technical arguments are needed to establish inequalities and estimates [3]. Then the resulting proofs are elementary but they tend to be relatively long and intricate [14,15].

The new proof given in the present note requires essentially no effort: it is elementary and requires no technical arguments; it is short and insightful. Its novelty consists in the easy observation that a certain limit is positive and that this implies that boundary points are excluded as minimizers of a certain penalty function. The remainder of the proof is obvious. All elementary proofs for the multiplier rule, including the one given in this note, make use of the theorem that a continuous function assumes its minimal value over a closed and bounded ball.

2. Proof of the Lagrange multiplier rule

Proof. By contraposition.

- *Preparations*. Assume that $\widehat{x}$ is an interior point of U that is a solution of the system of equations $f_i(x) = 0$, $1 \le i \le m$ and for which the n-dimensional row vectors $f_i'(\widehat{x})$, $0 \le i \le m$ are linearly independent. We note that this implies $n \ge m+1$. We will show that $\widehat{x}$ is not a minimum of the function $f_0(x)$ on the set of $x \in U$ for which $f_i(x) = 0$, $1 \le i \le m$.
  Without loss of generality it is assumed that $f_0(\widehat{x}) = 0, \widehat{x} = 0$ and $n = m+1$—the latter can be achieved by adding if necessary constraints $a_i x = 0$, $i \in I$, chosen in such a way that the set $a_i, i \in I$ completes the linearly independent set $f_i'(0)$, $0 \le i \le m$ to a basis of the space $(\mathbb{R}^n)^T$ of n-dimensional row vectors. Define $F(x) = (f_0(x), \ldots, f_m(x))^T$ for $x \in \mathbb{R}^n$. Then $F(0) = 0$, $F'(x)$ is the $n \times n$-matrix that has as rows the derivatives $f_i'(x)$, $0 \le i \le m = n-1$, F' is continuous and

$$F(x) = F'(0)x + o(|x|)(x \to 0). \tag{1}$$

  ('Landau little o')—as the f_i, $0 \le i \le m$ are continuously differentiable and $F(0) = 0$. The assumption that the vectors $f_i'(0)$, $0 \le i \le m = n-1$ are linearly independent means that the $n \times n$-matrix $F'(0)$ has rank $m+1 = n$, so $F'(0)$ is invertible. Choose for each $k \in \mathbb{N}$ a minimizer x_k of the penalty function

$$p_k(x) = |F(x) + (1/k^2, 0, \ldots, 0)^T| \tag{2}$$

  on the closed ball $|x| \le 1/k$, where $|\cdot|$ is the Euclidean norm.
- *Exclusion of boundary minimizers for the penalty function*. We consider the limit

$$L = \lim_{x \to 0} \frac{|F(x) + (|x|^2, 0, \ldots, 0)^T| - |(|x|^2, 0, \ldots, 0)^T|}{|F'(0)x|}. \tag{3}$$

  Note that

$$\lim_{x \to 0} |x|^2 / |F'(0)x| = 0.$$

  Therefore,

$$L = \lim_{x \to 0} |F(x)| / |F'(0)x| = 1 > 0, \tag{4}$$

  using (1) and the invertibility of $F'(0)$. It follows from (3) and (4) that for k sufficiently large and for each point $\bar{x}$ on the sphere $|x| = 1/k$, one has

$$|F(\bar{x}) + (|\bar{x}|^2, 0, \ldots, 0)^T| > |(|\bar{x}|^2, 0, \ldots, 0)^T|,$$

  that is, the penalty function p_k, given by (2), takes a larger value in $\bar{x}$ than in 0—here we use also $F(0) = 0$. Therefore, boundary points of the ball $|x| \le 1/k$ cannot be minimizers of the penalty function p_k, given by (2), on this ball. That is, $|x_k| < 1/k$.
- *Conclusion of the proof*. Moreover, for k sufficiently large, $|x_k| < 1/k$ implies that the matrix $F'(x_k)$ is invertible as $F'(0)$ is invertible and F' is continuous. From this, and as the minimizer x_k is an interior point of the closed ball $|x| \le 1/k$, it is obvious that for k sufficiently large

$$F(x_k) + (1/k^2, 0, \ldots, 0)^T = 0. \tag{5}$$

  Indeed, otherwise a local decrease of the penalty function p_k, given by (2), would be possible. Here is a precise analytic verification. The stationarity vector equation

$$\begin{aligned} 0 &= \frac{d}{dx} |F(x) + (1/k^2, 0, \ldots, 0)^T|^2 |_{x = x_k} \\ &= 2(F(x_k)^T + (1/k^2, 0, \ldots, 0)) F'(x_k) \end{aligned}$$

  holds as the minimizer x_k is interior. Here we use both the facts that the minimizers of a nonnegative valued function do not change if the objective function is squared, and that $|v|^2 = v^T v$ for a column vector v. This gives (5) as $F'(x_k)$ is invertible. That is, by the definition of F,

$$f_0(x_k) = -1/k^2 < 0 = f_0(0), \qquad f_i(x_k) = 0, \quad 1 \le i \le m.$$

  It follows that $\widehat{x} = 0$ is not a local minimizer of the function $f_0(x)$ on the set of points in U for which $f_i(x) = 0$, $1 \le i \le m$. This concludes the proof of the theorem.

Comment on the power of the rule. The power of the rule lies in the reversal of the natural order of two main tasks, elimination and differentiation. The natural order would be to eliminate m of the variables $x_1, \ldots, x_n$, using the equality constraints $f_i(x) = 0$, $1 \le i \le m$, substitute into the objective function $f_0(x)$, and then put the partial derivatives of the resulting expression in $n-m$ variables equal to zero. The elimination task is often a nonlinear problem

that is difficult or even impossible. Differentiating first gives the following statement, which is a reformulation of the conclusion of the theorem: all solutions $h \in \mathbb{R}^n$ of the system of linear equations

$$f_i'(\widehat{x})h = 0, \quad 1 \leq i \leq m \tag{6}$$

satisfy

$$f_0'(\widehat{x})h = 0,$$

provided the vectors $f_i'(\widehat{x})$, $1 \leq i \leq m$ are linearly independent. The remaining elimination task is now just a *linear* problem. It is easy to eliminate m of the n variables $h_1, \ldots, h_n$ using the system of linear equations (6). Then one can substitute in $f_0'(\widehat{x})h$. Finally one should put the coefficients of the resulting expression in $n-m$ variables equal to zero. This completes elimination without multipliers. In particular, multipliers are not essential to the power of the rule, but merely provide a convenient way to carry out the linear elimination task. See Ch 3 of [4] for more details on the power of the rule.

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